Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Directional vectors vs quantum state vectors

Files.

- File: `briefs/SuperpositionAndLinearity/SuperpositionAndLinearity.lean`
- Library: `SuperpositionAndLinearity`
- Imports: `ForPhyslib.QuantumMechanics.Qubit.Bloch`
- New file `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean` (imports `Mathlib.Analysis.InnerProductSpace.PiL2`, `Mathlib.LinearAlgebra.Matrix.DotProduct` and physlib's `Physlib.Relativity.PauliMatrices.Basic`): the Bloch vector of a vector of $\mathbb{C}^2$, the Born probability in terms of Bloch vectors, rays and orthogonality.

Setting. $H = \mathbb{C}^2$ (`EuclideanSpace ℂ (Fin 2)`), with $|z^+\rangle = e_0$, $|z^-\rangle = e_1$. The direction and the state vector are

$$v(\theta,\varphi) = (\cos\varphi\sin\theta,\ \sin\varphi\sin\theta,\ \cos\theta), \qquad \psi(\theta,\varphi) = \cos\tfrac{\theta}{2}\, e^{-i\varphi/2}\,|z^+\rangle + \sin\tfrac{\theta}{2}\, e^{i\varphi/2}\,|z^-\rangle\,,$$

(`direction`, `stateVector`). The Bloch vector of $\psi = \psi_0 e_0 + \psi_1 e_1$ is `blochVector` $\psi = (2\,\mathrm{Re}\,\bar{\psi}_0\psi_1,\ 2\,\mathrm{Im}\,\bar{\psi}_0\psi_1,\ |\psi_0|^2 - |\psi_1|^2)$, the vector of Pauli expectations $\langle\psi|\sigma_i|\psi\rangle$ (`blochVector_apply`). It is not normalized; its length is $\|\psi\|^2$.

Results.

Brief	Lean
eq. (1), $(v^x)^2 + (v^y)^2 + (v^z)^2 = 1$	`direction_dotProduct_self`
eqs. (2), (3), $p(z^+) = \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$	`prob_zp`
eqs. (2), (3), $p(z^-) = \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}$	`prob_zm`
eq. (3), $\langle\psi|\psi\rangle = 1$	`norm_stateVector`
abstract: $\psi(\theta,\varphi)$ is the state for the direction $v(\theta,\varphi)$	`blochVector_stateVector`: the Bloch vector of $\psi(\theta,\varphi)$ is $v(\theta,\varphi)$
Sec. 2, "linearity is in terms of probability"	`born_stateVector`: $\lvert\langle\psi(\theta',\varphi')\|\psi(\theta,\varphi)\rangle\rvert^2 = (1 + v'\cdot v)/2$; eq. (2) is the case $v' = e_z$ (`stateVector_zero_zero`, `direction_zero_zero`)
Sec. 2, $\theta + 2\pi$: negative components, same direction	`stateVector_add_two_pi`, `direction_add_two_pi`
Sec. 1, states are rays, not vectors	`exists_eq_smul_stateVector_iff`: two state vectors span the same ray if and only if they have the same direction
Sec. 2, the outcomes are opposite directions	`inner_stateVector_eq_zero_iff`: two state vectors are orthogonal if and only if their directions are opposite
abstract: every qubit state is of the form (3)	`exists_eq_smul_stateVector`: every unit vector is a phase times some $\psi(\theta,\varphi)$ with $0 \le \theta \le \pi$
(completeness)	`exists_direction_eq`: every unit vector of $\mathbb{R}^3$ is some $v(\theta,\varphi)$
In `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean`	
$\lvert\psi\rangle\langle\psi\rvert = (\|\psi\|^2 + a\cdot\sigma)/2$	`vecMulVec_eq`
Born probability	`norm_inner_sq`: $\lvert\langle\chi|\psi\rangle\rvert^2 = (\|\chi\|^2\|\psi\|^2 + a\cdot b)/2$
length of the Bloch vector	`blochVector_dotProduct_self`: $a\cdot a = \|\psi\|^4$
rays	`exists_eq_smul_iff`: unit vectors span the same ray iff their Bloch vectors agree
orthogonality	`inner_eq_zero_iff`: unit vectors are orthogonal iff their Bloch vectors are opposite

Differences from the brief.

- The brief relates the state vector to the direction through the probabilities along z only. The formalization uses the Bloch vector, which fixes the ray completely, so that "the state vector for a qubit, which also corresponds to a direction in physical space" becomes a theorem in

both directions (`blochVector_stateVector`, `exists_eq_smul_stateVector_iff`, `exists_eq_smul_stateVector`, `exists_direction_eq`).

- The brief says the probability "must be proportional to the component in that direction". It is proportional to the length $\overline{Az^-} = 1 + \cos\theta$ in Fig. 2, that is, affine in the component: $p(z^+) = (1 + v \cdot e_z)/2$. The formalization states the affine law for every pair of directions (`born_stateVector`).

- The bijection between rays of $\mathbb{C}^2$ and the unit sphere is proved pointwise (injectivity and surjectivity of the map from state vectors to directions); it is not stated as an equivalence or homeomorphism of the projective line with the sphere.

Problems found in the brief.

- The abstract and eq. (3) write the state as $\cos\frac{\theta}{2}e^{-i\varphi/2}\,|z^+\rangle + \sin\frac{\theta}{2}e^{i\varphi/2}\,|z^+\rangle$. The second ket is $|z^-\rangle$, as the computation of $p(z^-)$ in eq. (3) shows. With $|z^-\rangle$ the signs of the phases are the right ones: the Bloch vector is $v(\theta, \varphi)$ and not its mirror image $v(\theta, -\varphi)$.